On the unit $F \to \mathrm{R}\pi_*\pi^{-1}F$ for the birhizomic morphism of sites: the degree-zero isomorphism and the sheaf property

1 Statement

A *monoid* is an integral, saturated, sharp, commutative, unital semigroup, written additively, exactly as in the problem statement; it induces the translation invariant partial order $x \geq 0 \iff x \in M$ on M^{gp} and is *valuative* if that order is total. *Infinitesimal* maps, *birhizomicity at* a map, and *birhizomic covers* are used exactly as defined in the problem statement. For a monoid P: A_P is the category of monoid homomorphisms $P \to Q$; $B_P \subseteq A_P$ the full subcategory of the objects $P \to P[-c]^\sharp$; A_P^{op} carries the birhizomic topology, B_P^{op} the trivial topology; the inclusion $i : B_P^{\mathrm{op}} \subset A_P^{\mathrm{op}}$ induces $\pi^{-1} : \mathrm{Sh}(B_P^{\mathrm{op}}) \to \mathrm{Sh}(A_P^{\mathrm{op}})$ with right adjoint π_*.

Scope hypothesis. We prove the theorem for *finitely generated* P; from §2 on, "monoid" means finitely generated monoid wherever the finiteness of the face poset, Gordan's lemma, or pointed-cone separation is used (Proposition 2.1, Lemma 2.4, Corollary 2.5, and the face induction of §4). Note that every piece of every birhizomic cover of any object of A_P is *a priori* finitely presented over P (it is part of the definition of birhizomicity at a valuative target), hence finitely generated when P is; the f.g. hypothesis is used precisely at the places named above, and the general (non-finitely-generated) case is obstructed exactly there: $P \to P/K$ need not be finitely presented for arbitrary faces K, and the face poset can be infinite (see Remark 2.4).

We prove the degree-zero half of the desired quasi-isomorphism for finitely generated P, and we record precisely why the higher-degree half (and the general, non-finitely-generated case) remain open; see the Closing remark (after Proposition 4.20).

Theorem 1.1. *Let P be a* finitely generated *monoid* (*see the Scope hypothesis above*) *and F a presheaf of abelian groups on B_P^{op}. Then $\pi^{-1}F = G_F$ is a sheaf, and the degree-zero unit*

$$F \xrightarrow{\sim} \pi_*\pi^{-1}F = \mathrm{R}^0\pi_*\pi^{-1}F$$

is an isomorphism. (The higher direct images $\mathrm{R}^{q>0}\pi_*(\pi^{-1}F)$ are not *shown to vanish for arbitrary F; see the Closing remark.)*

Overview. The argument rests on a collapse of the birhizome to dimension one (§2): every valuative monoid is $\{0\}$ or $\mathbb{N}$ (Lemma 2.6); hence valuative points are functionals $P \to \mathbb{N}$ with half-space cones; and the birhizomic lifting condition at a nonzero target collapses to one test pair (Lemma 2.10, *atomicity*), giving the *normal form* (Theorem 2.12): every cover refines to one by *folds* ($P \to \mathbb{N}$) and *CFA pieces* ($P \to P/K$). §3 identifies $\pi^{-1}F$ with the explicit presheaf $G_F = F \circ B$ (Proposition 3.1) and proves it is a sheaf: separatedness is formal; the descent computation (Theorem 3.3) boils down to the fact that the overlap of any tight fold with the base piece $i(B(T))$ has ungrown kernel (Proposition 2.17, Example 2.16), so values are transported by identity transitions. §4 analyses the higher sheaf cohomology on $i(b)$: the probe covers (Definition 4.10) are cofinal, the kernel computations of the probe nerve (Lemma 4.11) split the Čech complex into

a fold strand and face strands, the fold terraces are killed (Lemma 4.17) and the dead sector is made coboundary-valued; the remaining residual computation is carried out as far as the structure of a general presheaf permits, and its limitation for arbitrary F is recorded honestly in the Closing remark.

Notation. For $T = (P \xrightarrow{\alpha_T} Q_T) \in A_P$: $\Phi_T := \ker(\alpha_T)$ (a face, Proposition 2.1(b)); $B(T) := (P \to P/\Phi_T) \in B_P$;

$$Z_T := \{x \in P^{\mathrm{gp}} : \alpha_T^{\mathrm{gp}}(x) \in Q_T\}, \qquad L_T := Z_T \cap (-Z_T),$$

the *relative saturation* and its *lineality*; $P \cap L_T = \Phi_T$ (Lemma 2.14). For a face K: P/K the face quotient (Proposition 2.1(a)); $K^{\mathrm{gp}} \subseteq P^{\mathrm{gp}}$ its generated group; $K^{\mathrm{gp}} \cap P = K$. A *functional* is $\varphi : M \to \mathbb{N}$; $[\varphi]$ its homothety class; $GF(\varphi)$ the positive generator of $\varphi^{\mathrm{gp}}(M^{\mathrm{gp}}) \subseteq \mathbb{Z}$; φ is *tight* on a sharp M if $\ker \varphi = \{0\}$. $\check{C}^\bullet, \check{H}^\bullet$ are alternating Čech complexes/cohomology.

2 Faces and the valuative collapse

2.1 Faces; the category B_P

Proposition 2.1. *Let M be a monoid, $K \subseteq M$ a face.*

(a) *$M/K := (M\backslash K)\cup\{0\}$, with 0 absorbing, is a monoid; $M \to M/K$ is initial among maps killing K; and $M/K \cong M[-c]^\sharp$ whenever c generates the face K; in particular $P[-c]^\sharp = P/\langle c\rangle_{\mathrm{face}}$, so B_P is the poset of faces of P: a morphism $P/K_1 \to P/K_2$ (unique) exists iff $K_1 \subseteq K_2$.*

(b) *Faces are exactly the sets $v^{-1}(0)$ for $v : M \to N$, N sharp; every face is f.g. as a monoid and they are finite in number.*

(c) *A face K is saturated relative to M: $K^{\mathrm{gp}} \cap M = K$. The face generated by $K_1 \cup K_2$ has group $K_1^{\mathrm{gp}} + K_2^{\mathrm{gp}}$.*

Proof. (a) $x = k_1 - k_2 \in M$, $k_i \in K$ gives $x + k_2 = k_1 \in K$, hence $x \in K$: so $M/K \hookrightarrow M^{\mathrm{gp}}/K^{\mathrm{gp}}$; the rest of the monoid axioms follow from the face property; initiality is clear. In $N := M[-c]^\sharp$ let e invert $\bar{c}$: $\bar{c} + e = 0$ and sharpness gives $\bar{c} = 0$: hence N receives the universal map from $M/\langle c\rangle$; conversely killing c inverts it (by 0): same universal problem. Poset claim: by initiality. (b) Kernels into sharp monoids are faces since $v(x) + v(y) = 0 \Rightarrow v(x) = v(y) = 0$ by sharpness; conversely $K = \ker(M \to M/K)$. A face is a supporting-hyperplane section of the f.g. rational polyhedral cone $\mathrm{Cone}(M)$: f.g. as a monoid (Gordan's lemma), and finitely many in number. (c) The first claim is the computation made at the start of (a). For the second: $\langle K_1 \cup K_2\rangle_{\mathrm{face}}$ sits inside the face $(K_1^{\mathrm{gp}} + K_2^{\mathrm{gp}}) \cap M$, whose group is $K_1^{\mathrm{gp}} + K_2^{\mathrm{gp}}$; conversely any $x \in (K_1^{\mathrm{gp}} + K_2^{\mathrm{gp}}) \cap M$ satisfies $x + k_1 + k_2 = k_1' + k_2'$ for some $k_i, k_i' \in K_i$ (write x as a difference and use $x \in M$), so x lies in the face generated by $K_1 \cup K_2$. □

Construction 2.2 (Pushouts in A_P)**.** For $T = (P \to Q_T)$, $X = (P \to Q_X)$, $Y = (P \to Q_Y)$ over T: with $\bar{P} := P/\Phi_T$ and $j_X : \bar{P} \to Q_X$, $j_Y : \bar{P} \to Q_Y$:

$$G := Q_X^{\mathrm{gp}} \oplus_{\bar{P}^{\mathrm{gp}}} Q_Y^{\mathrm{gp}} = \big(Q_X^{\mathrm{gp}} \times Q_Y^{\mathrm{gp}}\big)/\big\{(j_X^{\mathrm{gp}}(c), -j_Y^{\mathrm{gp}}(c)) : c \in \bar{P}^{\mathrm{gp}}\big\},$$

$W_0 := \mathrm{im}(Q_X \times Q_Y \to \breve{G})$, $\breve{G} := G/G_{\mathrm{tors}}$, and $X \otimes_T Y := \big(P \to (W_0^{\mathrm{sat}})^\sharp\big)$. Then $x \in \Phi_{X\otimes_T Y}$ iff $\alpha_X(x) \mapsto 0$ in $(W_0^{\mathrm{sat}})^\sharp$.

2.2 Finite presentation

Birhizomicity at a valuative target includes, by the problem's definition, the requirement that the map be *of finite presentation* (f.p.) in the category of monoids. We record what this means and that every map used as or inside a cover below satisfies it.

Proposition 2.3. *(i) A monoid map $f : M \to N$ is f.p. iff it presents N from M by finitely many generators and finitely many relations; equivalently N is a quotient of $M \oplus \mathbb{N}^r$ by a finitely generated congruence. (ii) For P f.g., a surjective $\varphi : P \to \mathbb{N}$ is f.p.: the congruence $\{(x, y) : \varphi(x) = \varphi(y)\}$ is a congruence on the f.g. monoid $P \times P$, generated as such by the kernel face relations and the analogous pointed-cone data; concretely it is the saturation of a f.g. congruence because the graph of $\ker \varphi^{\mathrm{gp}} \subseteq P^{\mathrm{gp}} \times P^{\mathrm{gp}}$ is a f.g. subgroup and P is saturated (this is the standard f.p.-of-integral-saturated-maps lemma; we give the argument in the next item). (iii) The CFA maps $\pi_K : P \to P/K$ are f.p. for every face K of the f.g. P: K is generated as a face by finitely many elements $k_1, \ldots, k_s$ (Proposition 2.1(b): f.g. as a monoid, and one may take a single element $c = k_1 + \cdots + k_s$), and $P[-c]$ is the localization, f.p.; the sharpening quotient is generated by the finitely many unit relations of a f.g. monoid, hence f.p. again. (iv) Pushouts of f.p. maps over a common map of monoids in our category are f.p.: they are quotients of finite colimits of f.g. monoids; saturation and sharpening are f.p. quotients of f.g. monoids.*

Proof. (i) Standard (Eilenberg-type exercise with congruences). (iii) π_K factors as $P \to P[-K] = P[-c] \to P/K$; each step is f.p. by the displayed presentations: $P[-c] = \langle P,\ e \mid \bar{c} + e = 0 \rangle$-style with one generator one relation; the sharpening of a f.g. monoid M is M modulo the f.g.-generated superdiagonal congruence $\{(u, -u) : u \in M^* \cup \text{f.g. generating set of units}\}$ (the unit group of a f.g. saturated sharp-quotient candidate is f.g. because it is the group of invertible elements of $\mathrm{Cone}(M)$, a finitely generated rational polyhedral cone). (ii) φ is the composite $P \to P/\ker\varphi \hookrightarrow \mathbb{N}$; the first is f.p. by (iii); the second exhibits $\mathbb{N}$ as $(P/\ker\varphi)$ with the finitely many relations generated by the image of any finite generating system of the f.g. group $P^{\mathrm{gp}}/\ker\varphi^{\mathrm{gp}}$ — the congruence determining the halfline inside the rank-one group is generated by the images of a finite basis of $\ker\varphi^{\mathrm{gp}}$ (basis finite by f.g. of P^{gp} and submodule finiteness over $\mathbb{Z}$): explicitly, choose $p_0 \in P$ with $\varphi(p_0) > 0$ minimal in the image group; then $\mathbb{N} = (P/\ker\varphi)/\!\sim$ with $\sim$ generated by $p \sim \varphi(p) \cdot [\bar{p}_0]$ for p in a generating set of P: f.p. (iv) The displayed Construction 2.2 exhibits $X \otimes_T Y$ from f.g. monoids by f.g. quotients (the $\bar{P}^{\mathrm{gp}}$-mixed congruence is generated by the images of a finite generating set of $\bar{P}^{\mathrm{gp}}$, f.g. when $\bar{P}$ is f.g.) and the saturation/sharpening steps of (iii). □

Remark 2.4 (Standing hypothesis P f.g.; uses in covers)**.** From here on, P is finitely generated. Consequently: every fold $\mathfrak{f}_\varphi$ and every CFA piece map used in any cover below is f.p. by Proposition 2.3(ii),(iii); the pushouts of Construction 2.2 among them are f.p. by (iv); the test maps (v/d) and $(d \cdot \mathrm{id}_{\mathbb{N}})$ of Lemma 2.10 are f.p. (targets $\mathbb{N}$); and every refinement map inserted below (Corollaries 2.18, 2.19, Lemma 4.18) is f.p. Without f.g. of P, the CFA family $\{P \to P/K\}$ may fail to be a cover (faces may fail to be finitely generated as faces, so π_K may fail to be f.p.), and the face poset can be infinite; these are the two obstructions to the immediate generalization of the argument to arbitrary P (cf. the Scope hypothesis in §1).

Lemma 2.5 (Covers may be taken surjective)**.** *Every cover in the slice admits a refinement whose piece maps are surjective monoid homomorphisms (in particular $Z_U = P + \Phi_U^{\mathrm{gp}}$ for every nerve piece, so Lemmas 4.11, 4.12 apply with the stated Z-hypotheses).*

Proof. A fold $\mathfrak{f}_\varphi$ with non-primitive $\varphi = d\varphi_0$ ($d = GF(\varphi)$, φ_0 primitive) is refined by the singleton piece $P \xrightarrow{\varphi_0} \mathbb{N}$: this covers $\mathfrak{f}_\varphi$: a valuative point factoring through $\mathfrak{f}_\varphi$ has the form $v = c\varphi_0$ with

$d \mid c$ (both functionals have a common primitive part), and then $v = (c/d) \circ \varphi_0$; the structure map φ_0 is birhizomic at the factor $(c/d\cdot) : \mathbb{N} \to \mathbb{N}$ by Lemma 2.10, the value groups $(c/d)\mathbb{Z}$ and $\big((c/d) \circ \varphi_0\big)^{\mathrm{gp}}(P^{\mathrm{gp}}) = (c/d)\mathbb{Z}$ coinciding since φ_0^{gp} is surjective. The new fold is surjective. CFA quotient maps are surjective by construction. Hence every cover admits a refinement—first the normal form Theorem 2.12, then the primitive reduction of its folds—whose piece maps are all surjective. For such U: $Z_U = \alpha_U^{\mathrm{gp},-1}(Q_U) = P + \Phi_U^{\mathrm{gp}}$ (any $q' = \alpha_U^{\mathrm{gp}}(z)$ lifts to $p' \in P$ by surjectivity, and $z = p' + (z - p') \in P + \ker \alpha_U^{\mathrm{gp}}$ with $\ker \alpha_U^{\mathrm{gp}} = \Phi_U^{\mathrm{gp}}$: Lemma 2.14). □

2.3 Valuative monoids

Lemma 2.6. *Every valuative monoid is* $\{0\}$ *or* $\mathbb{N}$.

Proof. Let $V \neq \{0\}$ be valuative; $\Gamma := V^{\mathrm{gp}}$ with convex chain $0 = H_0 \subsetneq \cdots \subsetneq H_s = \Gamma$.

Step 1. An archimedean ordered group order-embeds into $(\mathbb{R}, +)$ (Hölder: fix $a > 0$; $g \mapsto \sup\{m/n \in \mathbb{Q} : mg \leq na\}$); a f.g. subgroup of $\mathbb{R}$ is cyclic or dense.

Step 2: $s = 1$. If $s \geq 2$, fix $\gamma \in H_1$ positive and $\delta \in H_2 \setminus H_1$ positive. For every $n \geq 0$, $\delta - n\gamma \in V$: indeed its image in Γ/H_1 is $\bar{\delta} > 0$ in the induced total order. Set $\Lambda := H_2/H_1 \subseteq \Gamma/H_1$. The ordered group Λ has no proper nonzero convex subgroup (the preimage of such a subgroup in H_2 would be convex in Γ, hence equal to H_1 or H_2 by maximality of the chain), hence is archimedean; so by Step 1 it order-embeds in $(\mathbb{R}, +)$. Let G be a finite generating set of the monoid V, and put $c_0 := \min\{\bar{g} > 0 : g \in G,\ \bar{g} \in \Lambda\}$, where the minimum over the (finite) set of positive classes exists and is > 0 because some generator has positive class (otherwise $V \subseteq H_1$). Decompose $\delta - n\gamma = \sum_k g_k$ with $g_k \in G$: the Λ-classes of the summands are nonnegative with sum $\bar{\delta}$, so at most $N_0 := \mu_\Lambda(\bar{\delta})/c_0$ of them are outside H_1 (measured through the embedding $\mu_\Lambda : \Lambda \hookrightarrow \mathbb{R}$). Hence

$$\delta - n\gamma \;=\; \mathfrak{s}_n + h_n, \qquad \mathfrak{s}_n \text{ a sum of } \leq N_0 \text{ generators outside } H_1, \quad h_n \in V \cap H_1.$$

The $\mathfrak{s}_n$ range in a finite set; fix $\mathfrak{s}$ occurring for infinitely many n, and set $w := \delta - \mathfrak{s} \in \Gamma$. Then $w - n\gamma = h_n \in H_1$ for infinitely many n; taking a difference of two such indices shows $w \in H_1$, and $h_n = w - n\gamma \in V \cap H_1$ means $0 \leq n\gamma \leq w$ in the archimedean ordered group H_1 for infinitely many n: contradiction.

Step 3: Γ not dense. Suppose Γ is dense in $\mathbb{R}$; put $E := \mathbb{Q}\Gamma \subseteq \mathbb{R}$, write μ_E for the inclusion $E \hookrightarrow \mathbb{R}$, set $E^+ := \{e \in E : \mu_E(e) \geq 0\}$ and $d := \dim_{\mathbb{Q}} E$, and assume $d \geq 2$. Then $V = \Gamma \cap E^+$ and $C := \mathbb{Q}_{\geq 0} \cdot V$ is a full-dimensional rational polyhedral cone in E with $C \subseteq E^+$ (full-dimensional because $\Gamma = V - V$ spans E).

Lemma A: any $\mathbb{Q}$-rational form ℓ on E with $\ell \geq 0$ on V is zero. First, $\mu_E(V)$ is dense in $\mathbb{R}_{\geq 0}$: given $r \geq 0$, pick $\delta \in \Gamma$ with $0 < \mu_E(\delta) < 1/n$ (dense), and $m \in \mathbb{N}$ with $|m\mu_E(\delta) - r| < 1/n$ (archimedean division in $\mathbb{R}$); then $m\delta \in V$. Hence, viewing $E_{\mathbb{R}} = \mathbb{R}$ via μ_E and writing $\ell_{\mathbb{R}} := \ell \otimes \mathbb{R}$ (a continuous real form), the inequality $\ell_{\mathbb{R}} \geq 0$ holds on the dense subset $\mu_E(V)$ of $\mathbb{R}_{\geq 0}$, hence on all of $\mathbb{R}_{\geq 0}$; therefore $\ell_{\mathbb{R}} = c\,\mathrm{id}_{\mathbb{R}}$ with $c \geq 0$, i.e. $\ell(e) = c\,\mu_E(e)$ for all $e \in E$. If $c > 0$, the image $\mu_E(E) = c^{-1}\ell(E) \subseteq c^{-1}\mathbb{Q}$ would be a d-dimensional $\mathbb{Q}$-subspace of $\mathbb{R}$ inside a one-dimensional one: impossible as $d \geq 2$. Hence $c = 0$ and $\ell = 0$.

(3b) $C \neq E^+$. Write $C = \bigcap_{j=1}^r \{\ell_j \geq 0\}$ with $\mathbb{Q}$-rational ℓ_j (a rational polyhedral cone in E has rational supporting forms). If $C = E^+$, each $\ell_j \geq 0$ on $E^+ \supseteq V$: Lemma A gives $\ell_j = 0$ for all j, i.e. $C = E$: contradicting $C \subseteq E^+ \subsetneq E$.

(3c) Conclusion of Step 3. By (3b) there is $v \in E^+ \setminus C$; strict separation of the closed cone C from the point v inside the $\mathbb{Q}$-vector space (finite dimensional: standard strict separation applied rationally) yields a $\mathbb{Q}$-rational form ℓ with $\ell|_C \geq 0$ and $\ell(v) < 0$. Since $V \subseteq C$, $\ell|_V \geq 0$: Lemma

A gives $\ell = 0$, contradicting $\ell(v) < 0$. Therefore $d = 1$: Γ is a f.g. subgroup of $\mathbb{Q}$, hence cyclic: $\Gamma = \mathbb{Z}\gamma$, and $V = \mathbb{N}\gamma \cong \mathbb{N}$. □

Corollary 2.7. *Valuative points are* 0 *or functionals* $\varphi : M \to \mathbb{N}$*, with kernel face* $\{\varphi = 0\}$ *and half-space cone* $\{\varphi^{\mathrm{gp}} \geq 0\}$*. On a sharp* $M \neq \{0\}$ *tight functionals exist (strictly positive* $\mathbb{Q}$*-formals on the pointed cone* $\mathrm{Cone}(M)$*, denominator-cleared).*

2.4 Infinitesimal maps

Lemma 2.8. *If* $p : M \to N$ *is infinitesimal with* N *sharp then* $\ker p = \{0\}$*; if* N *is moreover valuative then* p^{gp} *is injective.*

Proof. As usual $x \prec y$ means $my \leq x \leq ny$ for some $m, n \in \mathbb{Z}$. First: let $x \in M$, $p(x) = 0$; with $X := -x$, $y := x$: $X \prec y$; $p(X) = 0 \in N$; $p(y) \prec 0$ reads $0 \leq p(y) \leq 0$ ✓; the condition gives $-x \in M$, a unit: $x = 0$. Second: $X = x_1 - x_2 \in \ker p^{\mathrm{gp}}$; $y := x_1 + x_2$: $X \prec y$; $p(X) = 0$; $p(y) \prec 0$ gives $p(x_1) + p(x_2) = 0$ in N; N sharp: each $= 0$; first part: $x_i = 0$. □

Lemma 2.9. *For* $p : M \to \mathbb{N}$ *nonzero:* p *infinitesimal iff* p^{gp} *injective and* M *conical:* $M = \{x \in M^{\mathrm{gp}} : p^{\mathrm{gp}}(x) \geq 0\}$*; then* $\pi := p/GF(p) : M \cong \mathbb{N}$*.*

Proof. "If" is immediate from the premise $p(x) \in \mathbb{N}$ alone. "Only if": $\ker p^{\mathrm{gp}} = 0$ by Lemma 2.8; for conicality let $p^{\mathrm{gp}}(x) > 0$, write $x = x_1 - x_2$, $y := x_1 + x_2$: $x \prec y$; $p(y) \prec p(x)$: $\mathbb{N}$ is archimedean: $p(y) \leq kp(x)$ for some k: premises OK: $x \in M$. □

2.5 Atomicity

Lemma 2.10 (Atomicity)**.** *Let* $f : Q \to Q'$ *be f.p. (this is part of birhizomicity; all our applications have it by Proposition 2.3) and* $\psi : Q' \to \mathbb{N}$ *nonzero. Then* f *is birhizomic at* ψ *iff* $\psi^{\mathrm{gp}}(Q'^{\mathrm{gp}}) = (\psi f)^{\mathrm{gp}}(Q^{\mathrm{gp}})$*.*

Proof. Necessity. With $v := \psi f$, $d := GF(v)$, test the pair $(h, p) := (v/d,\ d \cdot \mathrm{id}_{\mathbb{N}})$ (infinitesimal by Lemma 2.9; both maps are f.p. by Proposition 2.3(ii)/(iv), and the source $M = \mathbb{N}$ is f.g.); a lift u satisfies $du = \psi$, hence $\psi^{\mathrm{gp}}(Q'^{\mathrm{gp}}) \subseteq d\mathbb{Z} = (\psi f)^{\mathrm{gp}}(Q^{\mathrm{gp}})$; the other inclusion is automatic. *Sufficiency.* Given (h, p): $GF(p)$ divides $GF(v)$ ($v^{\mathrm{gp}} = p^{\mathrm{gp}} h^{\mathrm{gp}}$), so $\psi^{\mathrm{gp}}(Q'^{\mathrm{gp}}) = GF(v)\mathbb{Z} \subseteq GF(p)\mathbb{Z}$; define $u^{\mathrm{gp}} := (\pi^{\mathrm{gp}})^{-1} \circ (\psi^{\mathrm{gp}}/GF(p))$ where $\pi := p/GF(p)$ (isomorphism, Lemma 2.9); u^{gp} maps Q' into M by conicality and $p^{\mathrm{gp}} u^{\mathrm{gp}} = \psi^{\mathrm{gp}}$ (so $pu = \psi$ on Q': on the monoid: $p^{\mathrm{gp}} u^{\mathrm{gp}}(q') = \psi^{\mathrm{gp}}(q') \in \mathbb{N}$ ✓); and $uf = h$: $p^{\mathrm{gp}} u^{\mathrm{gp}} f^{\mathrm{gp}} = \psi^{\mathrm{gp}} f^{\mathrm{gp}} = p^{\mathrm{gp}} h^{\mathrm{gp}}$, p^{gp} injective; uniqueness: same injectivity. □

Corollary 2.11 (CFA pieces and covers)**.** *Let* Q *be a monoid and* K *a face. (a)* $Q \to Q/K$ *is birhizomic at every* $g : Q/K \to V$*,* V *valuative. (b)* $\{Q \to Q/K : K \text{ face of } Q\}$ *is a birhizomic cover (the* CFA cover*).*

Proof. (a) Nonzero $g = \psi$: Lemma 2.10: value-group equality holds since π_K^{gp} is surjective. Zero g: $ph = 0$ and Lemma 2.8 give $h = 0$; unique lift. (b) At v: $v = 0$: piece $K = Q$ with (a); $v = \varphi \neq 0$: piece $K = \ker \varphi$ with $\bar{\varphi}$ and (a). □

Theorem 2.12 (Normal form)**.** *Let* $T = (P \to Q_T)$*. Every birhizomic cover of* T *admits a refinement whose pieces are fold pieces* $\mathfrak{f}_\varphi = (P \xrightarrow{\varphi} \mathbb{N})$ *with* $\varphi(\Phi_T) = 0$*, and CFA pieces* $(P \to P/K)$ *with* $K \supseteq \Phi_T$*.*

Proof. For each piece $S = (P \xrightarrow{\alpha} Q)$ of the cover, the family

$$\mathcal{N}(S) = \{\mathfrak{f}_{\theta\alpha} : \theta : Q \to \mathbb{N}\} \ \cup \ \{Q/\mathfrak{k} : \mathfrak{k} \text{ face of } Q\}$$

is a cover of S: indeed, let $v : P \to V$ factor through α (these are the points covers of S are responsible for). If $V = \{0\}$: use $\mathfrak{k} = Q$. If $v = \theta\alpha$, $\theta : Q \to \mathbb{N}$ nonzero: write $\theta = d\theta_1$ with $GF(\theta_1) = 1$; then $v = (d\cdot) \circ (\theta_1\alpha)$ through the fold piece $\mathfrak{f}_{\theta_1\alpha}$; birhizomicity of the structure map θ_1 at $g = d \cdot \mathrm{id}$: Lemma 2.10: the condition $g^{\mathrm{gp}}(\mathbb{Z}) = (g\theta_1)^{\mathrm{gp}}(Q^{\mathrm{gp}})$ reads $d\mathbb{Z} = \theta^{\mathrm{gp}}(Q^{\mathrm{gp}}) = d\mathbb{Z}$ ✓. Compose these covers over all pieces (transitivity). □

Remark 2.13. In the slice topology, covers of T only need to answer those $v : P \to V$ that factor through $\alpha_T : P \to Q_T$ — in particular $v(\Phi_T) = 0$ and v factors through $P \to P/\Phi_T$: the problem's definition applied to the object α_T quantifies over $v = Q_T \to V$, and such v kill the kernel of α_T; all our covering constructions (Theorem 2.12, Corollaries 2.11, 2.18, 2.19) answer such v by structure maps killing Φ_T *because* the pieces used (folds with $\varphi(\Phi_T) = 0$, quotients P/K with $K \supseteq \Phi_T$) receive α_T (they lie over T).

2.6 Pushout kernels against folds

Fix T, $J := \Phi_T$, $\bar{P} := P/J$.

Lemma 2.14. *For U over T: $\alpha_U^{\mathrm{gp}}(L_U) = \{0\}$ and $P \cap L_U = \Phi_U$; the induced $Z_U/L_U \hookrightarrow Q_U$ is injective with image the sharp saturated submonoid $Q_U \cap \alpha_U^{\mathrm{gp}}(P^{\mathrm{gp}})$; the group J^{gp} satisfies $J^{\mathrm{gp}} \subseteq \ker \alpha_U^{\mathrm{gp}}$.*

Proof. $\lambda \in L_U$: $\pm\alpha_U^{\mathrm{gp}}(\lambda) \in Q_U$; sharp: $= 0$. Hence $x \in P \cap L_U \Rightarrow \alpha_U(x) = 0$: $P \cap L_U \subseteq \Phi_U$; conversely $\Phi_U \subseteq L_U$ directly. Injectivity of $Z_U/L_U \to Q_U$: $\alpha_U(z_1 - z_2) = 0$ with $z_i \in Z_U$: then $\pm(z_1 - z_2) \in Z_U$: $z_1 - z_2 \in L_U$. For J: $\alpha_U(J) = 0$. □

Lemma 2.15 (Kernel criterion)**.** *Let U over T and $\mathfrak{f} := \mathfrak{f}_\varphi$ with $\varphi(J) = 0$; $W := U \otimes_T \mathfrak{f}$. For $x \in P$:*

$$x \in \Phi_W \iff x \in \Phi_U \quad \text{or} \quad \exists\, t \geq 1,\, a \in P^{\mathrm{gp}} :\ a - tx \in Z_U,\ \varphi^{\mathrm{gp}}(a) \leq 0.$$

Proof. Reduce modulo J; write $j : \bar{P} \to Q_U$. By Construction 2.2, $\bar{x}$ dies in W iff there exist $t \geq 1$ and a unit u of W_0^{sat} with $[(\alpha_U(tx), 0)] + u = 0$ in G.

Necessity: the pair (A)+(B). Suppose $x \in \Phi_W$; if $x \in \Phi_U$ there is nothing to add. Otherwise take (t, u) as above. Writing the equality out with a class $u = [(\xi, m)]$: there is $c \in \bar{P}^{\mathrm{gp}}$ with $\alpha_U(tx) + \xi = j(c)$ and $m = -\varphi^{\mathrm{gp}}(c)$; lift c to $a_0 \in P^{\mathrm{gp}}$ and set $z_0 := a_0 - tx$: then $\alpha_U^{\mathrm{gp}}(z_0) = \xi$. The condition $+u \in W_0^{\mathrm{sat}}$ means that a positive multiple su has a representative in $Q_U \times \mathbb{N}$; raw representatives of su are $(\alpha_U^{\mathrm{gp}}(sz_0), -s\varphi^{\mathrm{gp}}(a_0))$, measured up to slides $(\xi, m) \mapsto (\xi + j(d), m - \varphi^{\mathrm{gp}}(d))$, $d \in \bar{P}^{\mathrm{gp}}$: hence for some d: $\alpha_U^{\mathrm{gp}}(sz_0 + d) \in Q_U$ and $s\varphi^{\mathrm{gp}}(a_0) + \varphi^{\mathrm{gp}}(d) \leq 0$. Take $(t', a) := (st,\ sa_0 + \hat{d})$ with $\hat{d}$ lifting d; then $a - t'x = sz_0 + \hat{d} \in Z_U$ (condition (A)) and $\varphi^{\mathrm{gp}}(a) \leq 0$ (condition (B)): the required pair exists.

Sufficiency. Conversely, given (t, a) with (A),(B): set $z := a - tx$, $\xi := \alpha_U^{\mathrm{gp}}(z) \in Q_U$, $m := -\varphi^{\mathrm{gp}}(a) \geq 0$, and $u := [(\xi, m)] \in W_0$; the relation $a = tx + z$ gives $[(\alpha_U(tx), 0)] + u = [(j(\bar{a}), -\varphi^{\mathrm{gp}}(\bar{a}))] = 0$ in G; hence $[(\alpha_U(tx), 0)] = -u$ lies in W_0 as well, so u is a unit, $t\bar{x}$ dies after sharpening, and x dies (saturation): $x \in \Phi_W$. That $x \in \Phi_U \Rightarrow x \in \Phi_W$ is clear (the structure map of W factors through that of U). □

Example 2.16 (Fold vs fold)**.** $U = \mathfrak{f}_\psi$, ψ tight on $\bar{P}$, and $\mathfrak{f}_\varphi$: the criterion (with $Z_U = \{\psi^{\mathrm{gp}} \geq 0\}$): if $[\varphi] = [\psi]$: (B)$\Rightarrow \psi(a) \leq 0$, (A): $\psi(a) \geq t\psi(x) \geq 1$ for $x \neq 0$: impossible: $\Phi_W = J$. If $[\varphi] \neq [\psi]$: the open rational wedge $\{\varphi \leq 0 < \psi\}$ in $\bar{P}^{\mathrm{gp}} \otimes \mathbb{R}$ is nonempty; scale a in it with $\psi(a) \geq t\psi(x)$: every $x \neq 0$ in $\bar{P}$ dies: $\Phi_W = \bar{P}$.

Proposition 2.17 (Equiface probes)**.** *Let U over T and $\varphi : P \to \mathbb{N}$ satisfy (G): $\varphi(L_U) = 0$ and $\varphi(z) > 0$ for $z \in Z_U \setminus L_U$ (such φ exist, and may be taken* primitive*: $Z_U/L_U \hookrightarrow Q_U$ is a sharp f.g. monoid by Lemma* 2.14*, so a strictly positive functional on it exists by Corollary* 2.7*; dividing out the content preserves the vanishing and positivity pattern; then automatically $\varphi(J) = 0$ since $J^{\mathrm{gp}} \subseteq L_U$). Then: (i) $\Phi_{\mathfrak{f}_\varphi} = \Phi_U$; (ii) $\Phi_{U \otimes_T \mathfrak{f}_\varphi} = \Phi_U$. In particular, for the base piece $iB(T)$ (where $Z = P + J^{\mathrm{gp}}$, $L = J^{\mathrm{gp}}$), every tight functional φ of $\bar{P}$ satisfies (G), so for every tight fold $\mathfrak{f}$: $\Phi_{\mathfrak{f}} = J$ and $\Phi_{\mathfrak{f} \otimes_T iB(T)} = J$.*

Proof. (i) $\Phi_{\mathfrak{f}_\varphi} = \{x \in P : \varphi(x) = 0\}$: $x \in P \subseteq Z_U$ with φ-zero: (G): $x \in L_U$: $x \in P \cap L_U = \Phi_U$ (Lemma 2.14) ✓. (ii) Lemma 2.15: suppose $x \notin \Phi_U$ dies: (t, a): $z := a - tx \in Z_U$, $\varphi(a) \leq 0$. If $\varphi(z) < 0$: but $z \in Z_U$ and (G) gives $\varphi(z) \geq 0$ (with > 0 off L_U): contradiction. So $\varphi(z) \geq 0$; then $\varphi(z) = \varphi(a) - t\varphi(x)$ gives $\varphi(a) \geq t\varphi(x)$; with $\varphi(a) \leq 0$ and $\varphi(x) > 0$ (as $x \in P \subseteq Z_U$, $x \notin L_U$ since $x \notin \Phi_U = P \cap L_U$): $0 \geq \varphi(a) \geq t\varphi(x) \geq 1$: contradiction. Hence $\Phi_W = \Phi_U$ ✓. The $iB(T)$-computation: $Z_{iB(T)} = P + J^{\mathrm{gp}}$, $L_{iB(T)} = J^{\mathrm{gp}}$ ✓(sharpness of $\bar{P}$: $\pm\bar{x} \in \bar{P} \Rightarrow \bar{x} = 0$, i.e. $x \in J$); (G) for tight φ: $\varphi(J^{\mathrm{gp}}) = 0$ ✓; $z \in Z \setminus L$ has $\bar{z} \neq 0$, so $\varphi(\bar{z}) > 0$ ✓. □

Corollary 2.18 (Probes exist in covers)**.** *Let $R = \{T \to T_i\}$ be a cover of T, and let $\varepsilon : P \to \mathbb{N}$ be any functional* coming from an equiface probe*, i.e. satisfying for some piece U over T the condition (G) of Proposition* 2.17 (*in particular $\Phi_{\mathfrak{f}_\varepsilon} = \Phi_T$ by (i) loc. cit.*). *Then ε may be taken primitive (replace by its primitive part), and R admits a refinement containing $\mathfrak{f}_\varepsilon$. In particular (with $U = iB(T)$, where every tight ε works): every cover admits a refinement containing a tight fold with kernel exactly Φ_T.*

Proof. By Proposition 2.17, equiface probes may be chosen primitive: replace the given ε by its primitive part, which satisfies the same vanishing and positivity conditions, hence also defines a fold with kernel Φ_T. Apply the cover property of R at the point $v := \varepsilon$: some member T_i admits $\theta : Q_{T_i} \to \mathbb{N}$ with $\theta\alpha_i = v$ and $T \to T_i$ birhizomic at θ; by Lemma 2.10, $\theta^{\mathrm{gp}}(Q_{T_i}^{\mathrm{gp}}) = v^{\mathrm{gp}}(P^{\mathrm{gp}})$, i.e. $GF(\theta) = GF(v) = 1$. Refine T_i by its CFA cover (Corollary 2.11(b)) and keep the member over the face $\ker\theta$: the induced $\bar{\theta} : Q_{T_i}/\ker\theta \hookrightarrow \mathbb{N}$ is injective with saturated sharp GF=1-image in $\mathbb{N}$, hence an isomorphism; the member $S := (P \to Q_{T_i}/\ker\theta)$ is therefore isomorphic over P to $\mathfrak{f}_\varepsilon$. (All maps used are f.p.: the slice maps by Proposition 2.3(ii)–(iii), and the refinement composites by (iv) therein.) The composed refinement of R contains $\mathfrak{f}_\varepsilon$. □

Corollary 2.19 (Probe covers)**.** *Let $\bar{P} \neq \{0\}$ and let $\chi : \bar{P} \to \mathbb{N}$ be a primitive tight functional. Then the family*

$$\square_1 \;:=\; \{\mathfrak{f}_{\chi\bar{\pi}}\} \;\cup\; \{(P \to P/K) : K \supseteq J \text{ face}\}$$

(one tight primitive fold on the ray $[\chi]$, together with all CFA pieces) already covers T.

Proof. Let $v = \psi\bar{\pi}$, $\psi : \bar{P} \to \mathbb{N}$ nonzero; $\ker\psi =: K^0 \supseteq 0$ a face of $\bar{P}$; let $K :=$ preimage of K^0 in P (a face $\supseteq J$). Use the piece P/K: v factors $P/K \xrightarrow{\bar{\psi}} \mathbb{N}$; birhizomicity of the structure map $P \to P/K$ at $\bar{\psi}$: Corollary 2.11(a). If $\ker\psi = 0$ and $[\psi] = [\chi]$: $\psi = d\chi$: use the fold piece and Lemma 2.10: requires $d\mathbb{Z} = (\psi\bar{\pi})^{\mathrm{gp}}(P^{\mathrm{gp}}) = d\mathbb{Z}$ ✓. The zero point: dead piece. □

3 The inverse image is a sheaf

Proposition 3.1. *$i : B_P \to A_P$ has right adjoint $\varphi_0 : A_P \to B_P$, $T \mapsto B(T)$; hence the pointwise left Kan extension of F along i is $G_F := F \circ \varphi_0$, and $G_F(i(b)) = F(b)$.*

Proof. $\mathrm{Hom}_{A_P}(i(P \to P/K), T)$ is a singleton iff $K \subseteq \Phi_T$, else empty (Proposition 2.1), matching $\mathrm{Hom}_{B_P}((P \to P/K), B(T))$: adjunction. Kan extension along a functor with a right adjoint is precomposition with the adjoint; pointwise: $(i \downarrow T)$ has terminal object $B(T)$. □

Proposition 3.2 (Separatedness)**.** *For every cover R of T, $G_F(T) \to \prod_{T_i \in R} G_F(T_i)$ is injective.*

Proof. By Theorem 2.12 and Corollary 2.18, R admits a refinement (pulling back along which the map in question factors) containing a tight fold $\mathfrak{f}$ with $\Phi_\mathfrak{f} = J$; the restriction $G_F(T) \to G_F(\mathfrak{f})$ is $F(\mathrm{id}_{B(T)})$: injective. □

Theorem 3.3 (Descent)**.** *For every cover R of T, $G_F(T) \xrightarrow{\sim} \check{H}^0(R; G_F)$. Hence G_F is a sheaf on $(A_P^{\mathrm{op}})_{\mathrm{birhiz}}$; consequently $\pi^{-1}F = G_F$ and $F(b) \xrightarrow{\sim} (\pi_*\pi^{-1}F)(b)$, canonically.*

Proof. Injectivity is Proposition 3.2. Let $(s_U) \in \check{H}^0(R; G_F)$. By Theorem 2.12, Corollary 2.18, and Corollary 2.19, we may refine (and pull the system back; this does not change its fate for surjectivity) so that R contains: a tight fold $\mathfrak{f}_\star$; the base piece $iB := i(B(T))$; all CFA pieces $(P \to P/K)$, $K \supseteq J$; and for every piece U of R a (G)-probe $\mathfrak{f}_{\varepsilon_U}$ with ε_U satisfying (G) for U (Proposition 2.17). The nerve of the refined cover contains all pushouts of its pieces. Set

$$s^0 \;:=\; s_{iB} \in F(B(T)).$$

Claim A: $s_\star = s^0$. The kernel of $W := \mathfrak{f}_\star \otimes_T iB$ is J (Proposition 2.17, iB-case: every tight functional satisfies (G) for iB). Compatibility at W: the two restriction maps $G_F(\mathfrak{f}_\star) \to G_F(W)$ and $G_F(iB) \to G_F(W)$ are both $F(\mathrm{id}_{B(T)})$ (as $\Phi_\star = \Phi_W = \Phi_{iB} = J$); hence $s_\star = s_{iB} = s^0$.

Claim B: for every piece U of R, $s_U = F(B(T) \twoheadrightarrow B(U))(s^0)$. Case $\Phi_U = J$. The probe piece $\mathfrak{f}_{\varepsilon_U}$ has kernel J (Prop. 2.17(i)), i.e. ε_U is tight on $\bar{P}$, hence satisfies (G) for iB (Prop. 2.17, last clause); so $\Phi_{\mathfrak{f}_{\varepsilon_U} \otimes_T iB} = J$ and the two restriction maps into the overlap are $F(\mathrm{id}_{B(T)})$: compatibility gives $s_{\varepsilon_U} = s_{iB} = s^0$. Also, by Prop. 2.17(ii), $U \otimes_T \mathfrak{f}_{\varepsilon_U}$ has kernel $\Phi_U = J$ with identity transitions from both factors, so $s_U = s_{\varepsilon_U} = s^0$, the required form since $B(U) = B(T)$. *Case $\Phi_U = K \supsetneq J$.* The probe piece $\mathfrak{f}_{\varepsilon_U}$ has kernel K (Prop. 2.17(i)), so $U \otimes \mathfrak{f}_{\varepsilon_U}$ has kernel K (loc. cit. (ii)) with identity transitions: $s_U = s_{\varepsilon_U}$. We now compute the kernel of $W := \mathfrak{f}_{\varepsilon_U} \otimes_T iB$ via Lemma 2.15 ($Z_{iB} = P + J^{\mathrm{gp}}$; write bars for reduction mod J, so Z_{iB}-membership is "$\overline{\cdot} \in \bar{P}$", and $\varepsilon_U = \psi\bar{\pi}$ with $\psi : \bar{P} \to \mathbb{N}$ whose kernel is exactly the face K/J): *claim: $\Phi_W = K$.*

If $\bar{x} \in K/J$: take $t := 1$ and a lift a of $\bar{a} := \bar{x}$; then $a - x$ reduces to 0, hence lies in $J^{\mathrm{gp}} \subseteq P + J^{\mathrm{gp}} = Z_{iB}$: (A) holds; and $\varepsilon_U(a) = \psi(\bar{x}) = 0$: (B) holds. Hence $x \in \Phi_W$, i.e. $K \subseteq \Phi_W$.

Conversely let $x \in \Phi_W$ via (t, a): (B): $\psi(\bar{a}) \leq 0$, with $\bar{a} = \overline{a - tx} + t\bar{x}$, where $\bar{z} := \overline{a - tx} \in \bar{P}$ by (A). Two cases: if $\psi(\bar{a}) = 0$, then $\bar{a} \in K/J$; the equation $t\bar{x} = \bar{a} - \bar{z}$ with $\bar{z} \in \bar{P}$ is an equality $t\bar{x} + \bar{z} = \bar{a}$ inside $\bar{P}$, i.e. $t\bar{x}$ and $\bar{z}$ are elements of $\bar{P}$ whose sum lies in the face $K/J \subseteq \bar{P}$; by the face property, $t\bar{x} \in K/J$, whence $\bar{x} \in K/J$ (faces are saturated; Proposition 2.1(c)), i.e. $x \in K$. If $\psi(\bar{a}) < 0$: then $t\psi(\bar{x}) = \psi(\bar{a}) - \psi(\bar{z}) \leq \psi(\bar{a}) < 0$ while $\psi(\bar{x}) \geq 0$: impossible. Hence $\Phi_W \subseteq K$: equality. Compatibility at W: $G_F(\mathfrak{f}_{\varepsilon_U}) \to G_F(W)$ is $F(\mathrm{id}_{P/K})$ while $G_F(iB) \to G_F(W)$ is $F(B(T) \twoheadrightarrow P/K)$; hence $s_{\varepsilon_U} = F(B(T) \twoheadrightarrow P/K)(s^0)$, and with $s_U = s_{\varepsilon_U}$ above, Claim B follows.
Endgame. Claim A+B: the pulled system on the final refinement R' has the form $\iota_{R'}(s^0)$ with $s^0 \in F(B(T)) = G_F(T)$: it lies in the image of $G_F(T)$. Hence every class of $\mathrm{colim}_R\, \check{H}^0(R; G_F) = G_F^+(T)$

lies in the image of $G_F(T)$, so $G_F(T) \to G_F^+(T)$ (injective by Proposition 3.2) is an isomorphism. For the original cover R, consider

$$G_F(T) \longrightarrow \check{H}^0(R; G_F) \longrightarrow G_F^+(T) :$$

the composite is bijective and the first map is surjective (a constant system is compatible), so the second map is injective; both are therefore bijective: $\check{H}^0(R; G_F) = G_F(T)$ on the nose. G_F is then a sheaf (descent on all covers: glue maps iso), $\pi^{-1}F = G_F$, and $(\pi_*\pi^{-1}F)(b) = G_F(i(b)) = F(b)$ (Proposition 3.1). □

4 Higher direct images

Let $b = (P \to P/J)$, $\bar{P} = P/J$, $\mathcal{G} := G_F = \pi^{-1}F$ (a sheaf by Theorem 3.3). We now compute $H^q(i(b), \mathcal{G})$.

Definition 4.1 (The Čech nerve of a cover)**.** Let $R = \{i(b) \to T_i\}$ be a cover of $X := i(b)$ in the sliced site. Its *nerve* is the simplicial object of the slice whose degree q part is the disjoint family of the iterated pushouts $\overline{\tau} := \overline{U_0} \cap \cdots \cap \overline{U_q} := U_0 \otimes_{i(b)} \cdots \otimes_{i(b)} U_q$ over alternating tuples $\tau = (U_0, \ldots, U_q)$ of pieces (the pushouts exist and their piece maps are finitely presented, by Construction 2.2 and Proposition 2.3(iv)); face maps are given by forgetting a factor, degeneracies by repeating one; the alternating Čech complex of a presheaf $\mathscr{W}$ is

$$\check{C}^q(R; \mathscr{W}) \;=\; \prod_{\tau \text{ alternating}} \mathscr{W}(\overline{\tau}), \qquad (\partial e)_\tau = \sum_{i=0}^{q+1} (-1)^i \, \mathscr{W}(d_i)\big(e_{\partial_i \tau}\big),$$

where $d_i : \overline{\tau} \to \overline{\partial_i \tau}$ is the structure map forgetting the i-th factor.

Lemma 4.2 (Structure maps between nerve pieces)**.** *For nerve pieces W_1, W_2 over $i(b)$ of covers whose pieces are folds or CFA-quotients (as in all covers used below): a structure map $W_1 \to W_2$ over X exists iff $\Phi_{W_1} \subseteq \Phi_{W_2}$ provided both structure maps are surjective; in general the surjective pieces among the nerve (iB, the $i(P/K)$, the folds satisfied) satisfy the criterion, and the kernel-equal case gives the identity transition.*

Proof. If the structure map of W_2 is a face quotient P/K_2 or the fold $\mathbb{N}$-projection, a map into it out of W_1 is the same as a factorization killing the kernel (universal property of P/K_2: Proposition 2.1(a); for the folds, maps between folds on the same ray are the coordinate projections). Killing Φ_{W_2} in W_1 is possible exactly when $\Phi_{W_1} \subseteq \Phi_{W_2}$. When the kernels agree, the induced map on B-levels is the identity of P/Φ, i.e. $F(\mathrm{id})$. □

Remark 4.3 (The comparison theorem used)**.** We use Verdier's hypercovering theorem (SGA 4, Exp. V, Théorème 7.4.1), in the following consequence: on a site with fibre products — the sliced birhizomic site has them by Construction 2.2 — the sheaf cohomology of an abelian sheaf $\mathscr{W}$ on X is computed by $H^q(X, \mathscr{W}) = \mathrm{colim}_R \, \check{H}^q(R; \mathscr{W})$, the colimit over all covers of the alternating Čech cohomology of their nerves as in Definition 4.1; and a class vanishes in $H^q(X, \mathscr{W})$ iff it pulls to zero along some refinement. (No other external input is used.)

4.1 The nerve complex: depth, strands, and transfers

Throughout, R is a cover of X in the sliced site, and all presheaves take values in abelian groups. For an alternating tuple $\tau = (U_0, \ldots, U_q)$ of pieces write $\overline{\tau} := U_0 \otimes_X \cdots \otimes_X U_q$ for its nerve piece (Definition 4.1), and $\partial_i \tau := (U_0, \ldots, \widehat{U_i}, \ldots, U_q)$.

Remark 4.4 (Normalization)**.** Definition 4.1 defined $\check{C}^\bullet(R;\mathscr{W})$ directly on *alternating* tuples. Equivalently one may start from the unreduced complex over all tuples and pass to the quotient by the degenerate subcomplex: the two compute the same cohomology (the degenerate cochains form an acyclic subcomplex: standard normalization of simplicial abelian groups, used here only as this identification). We work throughout with alternating tuples.

Definition 4.5 ($\mathscr{W}$-strand of a family)**.** For a subfamily $\mathcal{J}$ of pieces, the *$\mathcal{J}$-strand* is the subquotient of $\check{C}^\bullet$ obtained by restricting to tuples all of whose pieces lie in $\mathcal{J}$: it is a subcomplex when it is closed under ∂ (i.e. whenever every face of a tuple in $\mathcal{J}$, with its values transported along the structure map, still has its entries in $\mathcal{J}$); more generally, a family of strands is *differential-closed* when the differential of a cochain supported on it stays supported on it.

Lemma 4.6 (Constant complexes)**.** *Let $\mathcal{J}$ be a differential-closed family of pieces on which $\mathscr{W}$ takes a constant value A with all structure maps inducing* id_A*, and suppose $\mathcal{J}$ has a minimum piece H ($H \subseteq U$, hence $\overline{H} \hookrightarrow \overline{U}$ and the induced $\mathscr{W}$-map is* id_A*, for all $U \in \mathcal{J}$) so that every nerve piece of $\mathcal{J}$ receives a structure map from $\overline{H}$. Then the $\mathcal{J}$-strand computes $\check{H}^0 = A$ and $\check{H}^{q\geq 1} = 0$.*

Proof. For $q \geq 1$ define $(he)_\tau := e_{(H,\tau)}$ at alternating tuples τ of $\mathcal{J}$ (when (H,τ) has a repetition its entry is 0, consistent with the alternating convention). Since $\overline{(H,\tau)} = \overline{H} \otimes \overline{\tau}$ receives $\overline{H}$, the value $e_{(H,\tau)}$ is read off along the structure map out of the H-level, i.e. is an element of A. The face relations $\partial_0(H,\tau) = \tau$, $\partial_{i+1}(H,\tau) = (H, \partial_i\tau)$ give, with all structure maps inducing id_A,

$$(\partial he)_\tau = \sum_{i=0}^{q} (-1)^i e_{(H,\partial_i\tau)}, \qquad (h\partial e)_\tau = e_\tau - \sum_{i=0}^{q} (-1)^i e_{(H,\partial_i\tau)},$$

so $(\partial h + h\partial)e = e$: $\check{H}^{q\geq 1} = 0$. For $q = 0$: the cocycle condition at the pair (H,U) reads $e_U = \mathscr{W}(\overline{H} \hookrightarrow \overline{U})(e_H) = e_H$, so $\check{H}^0 = A$ by evaluation at H. □

Lemma 4.7 (Exact subquotient and signed transport)**.** *Let $\check{C}^\bullet = \check{C}' \oplus \check{C}''$ be a decomposition into differential-closed subcomplexes with $\check{C}'$ acyclic. Then $\check{H}^\bullet(\check{C}) = \check{H}^\bullet(\check{C}'')$; and if $z = (z', z'')$ is a degree-$q \geq 1$ cocycle, there is a cochain $w' \in \check{C}'^{\,q-1}$ with $z' = \partial w'$, so that $z - \partial w'$ is a cocycle supported on $\check{C}''$ whose $\check{C}''$-component differs from z'' by the signed transport $z'' - (\mathrm{pr}'' \circ \partial)w'$ (this component is a cocycle; if $\check{C}''$ is closed under ∂ on its own it equals z'').*

Proof. Acyclicity of $\check{C}'$ gives w' with $z' = \partial w'$ (the cocycle z restricts to a cocycle on $\check{C}'$ by differential-closedness). Then $z - \partial w' = (0,\, z'' - \mathrm{pr}''\partial w')$: a cocycle supported on $\check{C}''$, as stated; its cohomology class equals that of z, and the long exact sequence of $0 \to \check{C}'' \to \check{C} \to \check{C}' \to 0$ identifies $\check{H}^\bullet(\check{C})$ with $\check{H}^\bullet(\check{C}'')$ whenever $\check{C}'$ is acyclic. □

Lemma 4.8 (Depth of nerve pieces)**.** *Assign to each piece U over X its* depth*: the length of a maximal chain in the face poset of P from J up to the kernel face Φ_U (well-defined since the face poset is finite, Proposition 2.1(b); the actual constants never matter); the dead piece D (kernel P) has maximal depth, a kernel-J fold or iB depth 0. Depth is* order-respecting on the nerve*: forgetting a factor can only increase depth,* $\mathrm{depth}(\partial_i\tau) \leq \mathrm{depth}(\tau)$*, since $\overline{\partial_i\tau}$ receives a structure map from $\overline{\tau}$, whence $\Phi_{\overline{\tau}} \subseteq \Phi_{\overline{\partial_i\tau}}$ by Lemma 4.2.*

Proof. The map $\overline{\tau} \to \overline{\partial_i\tau}$ forgetting the i-th factor is a structure map over X; a structure map $W_1 \to W_2$ over X forces $\Phi_{W_1} \subseteq \Phi_{W_2}$ (Lemma 4.2). Both claims follow from monotonicity of the kernel face along structure maps and the definition of depth as poset-length of the kernel. □

Remark 4.9 (Homotopy transfer along refinements)**.** If R' refines R with a choice of refinement map $\sigma : R' \to R$ of families (each R'-piece factoring through its σ-image), pullback of cochains along σ is a cochain map on nerves; two choices of σ induce cochain homotopic maps (the standard comparison for Čech refinements; the homotopy inserts at position i the tuple $(\sigma U_0, \dots, \sigma U_{i-1}, U_i, \dots, U_q)$). These comparison maps assemble the colimit of Remark 4.3; in particular, a class vanishes in sheaf cohomology iff it pulls to zero along *some* refinement, and any cocycle-level identity established after a pullback persists in the colimit.

Definition 4.10. A cover of $X := i(b)$ is *standard* if all its piece maps are surjective (always achievable, Lemma 2.5), it is in the normal form of Theorem 2.12, and moreover it contains the pieces iB, $i(P/K)$ for all faces $K \supseteq J$, the dead piece D, and two kernel-J folds on distinct rays whenever $\dim \bar{P} \geq 2$ (two nonproportional strictly positive rational forms on the pointed cone $\mathrm{Cone}(\bar{P})$; when $\dim \bar{P} \leq 1$ no second fold is demanded, cf. the rank-one comment in Step 2 of Proposition 4.19). A standard cover is *rigid* if no two distinct pieces admit a structure map between them, except the quotient maps forced by inclusion of faces. Every cover admits a standard rigid refinement: first refine to normal form (Theorem 2.12), then intersect objectwise with the probe family $\{\mathfrak{f}\} \cup \{i(P/K)\} \cup \{iB\} \cup \{D\}$ of Corollary 2.19 (possibly doubled to two rays; the intersection is a cover, and all named pieces occur as their own intersections); then delete, inductively, any piece with a nonidentity structure map to another surviving piece—a point served by the deleted piece is served via the composition, whose birhizomicity at the point follows from the deleted map's birhizomicity at the target factorization. ("Refined normal cover" below means: standard rigid cover.)

Lemma 4.11 (Kernels in the probe nerve)**.** *In the nerve of $\square_b$: (i) $\Phi_{\mathfrak{f} \otimes P/K} = K$ for every face $K \supseteq J$; in particular the overlap $\mathfrak{f} \otimes iB$ has B-level b, and $\mathfrak{f} \otimes P/K$ is dead for $K \supsetneq J$; (ii) $\Phi_{P/K_1 \otimes P/K_2} = K_1 \vee K_2$; (iii) consequently every nerve piece's $\mathcal{G}$-value is F of a single face quotient, with canonical quotient maps between them.*

Proof. (ii): quotients compose; by Proposition 2.1(c) the joint face has group $K_1^{\mathrm{gp}} + K_2^{\mathrm{gp}}$ and $(K_1 \vee K_2)^{\mathrm{gp}} \cap P = K_1 \vee K_2$ refines the kernel: the pushout of $P/K_1 \leftarrow \bar{P} \to P/K_2$ is a quotient of both: kernel exact face $K_1 \vee K_2$ ✓. (i): Lemma 2.15 with $U = (P \to P/K)$, whence $Z_U = P + K^{\mathrm{gp}}$, $L_U = K^{\mathrm{gp}}$: the jump condition reads: some $t \geq 1$, $a \in P^{\mathrm{gp}}$ with $z := a - tx \in P + K^{\mathrm{gp}}$ and $\chi(a) \leq 0$. Write everything downstairs: $\bar{z} = \bar{a} - t\bar{x} \in \bar{P} + (K/J)^{\mathrm{gp}}$: $\bar{\chi}(\bar{a}) - t\bar{\chi}(\bar{x}) = \bar{\chi}(\bar{z})$; with $\bar{\chi}$ tight: $\bar{\chi}(\bar{a}) \leq 0$; decompose $\bar{a} = \bar{\omega} + \bar{\kappa}$ with $\bar{\omega} \in \bar{P}$ and $\bar{\kappa} \in (K/J)^{\mathrm{gp}}$ (possible since $\bar{a} \in t\bar{x} + \bar{P} + (K/J)^{\mathrm{gp}} \subseteq \bar{P} + (K/J)^{\mathrm{gp}}$). Then $0 \geq \bar{\chi}(\bar{a}) = \bar{\chi}(\bar{\omega}) + \bar{\chi}(\bar{\kappa})$ with $\bar{\chi}(\bar{\omega}) \geq 0$. Case A: $K = J$: $\bar{\kappa} = 0$: gives $\bar{\chi}(\bar{\omega}) = 0$: $\bar{\omega} = 0$ (tight): then $\bar{a} = 0$: jump condition was $-t\bar{x} \in \bar{P}$: sharpness: $\bar{x} = 0$: $x \in J$: so $\Phi = J$ ✓. Case B: $K \supsetneq J$: $(K/J)^{\mathrm{gp}} \neq 0$; since $\bar{\chi}$ is a group homomorphism, $\bar{\chi}((K/J)^{\mathrm{gp}}) = d\mathbb{Z}$ for some $d \geq 1$ (it is nonzero: otherwise $K^{\mathrm{gp}} \subseteq \ker \bar{\chi}^{\mathrm{gp}}$, hence $K \cap \bar{P} = K \subseteq \ker \bar{\chi} = 0$: contradiction with $K \neq J$). Pick $\bar{\kappa}_N \in (K/J)^{\mathrm{gp}}$ with $\bar{\chi}(\bar{\kappa}_N) = -Nd$. Then for ANY $\bar{x} \in \bar{P}$, with a defined by $\bar{a} := t\bar{x} + \bar{\kappa}_N$ (N large so $\bar{\chi}(\bar{a}) \leq 0$), we get $\bar{a} - t\bar{x} = \bar{\kappa}_N \in (K/J)^{\mathrm{gp}} \subseteq \bar{Z}$ ✓and $\bar{\chi}(a) \leq 0$ ✓: every x dies: the piece is dead ✓. (For the second clause of (i): $\mathfrak{f} \otimes iB$ is Case A.) □

Lemma 4.12 (Kernels of general nerve pieces)**.** (*a*) *The overlap of two kernel-J folds has kernel J if their functionals are proportional (same ray) and is dead otherwise.* (*b*) *Longer overlaps are governed by the two exact claims:*

(*C1*) *If $\Phi_U = J$ and $Z_U = P + J^{\mathrm{gp}}$ (as is the case for all pieces with surjective structure map, Lemma 2.14), and φ is tight on $\bar{P} = P/J$, then $\Phi_{U \otimes \mathfrak{f}_\varphi} = J$.*

(*C2*) *If* $\Phi_U = K$ *and* $Z_U = P + K^{\mathrm{gp}}$*, and* φ *vanishes exactly on the face* K *(tight on* P/K*), then* $\Phi_{U \otimes \mathfrak{f}_\varphi} = K$ *(applied in the iteration with* K *replaced by the finer join).*

Consequently: if two positions are kernel-J *folds on distinct rays, the piece is dead; otherwise its kernel equals the kernel of the sub-overlap of its non-fold positions, computed inductively by (C2).*

Proof. (a) Example 2.16. (C1) By Lemma 2.15 a jump requires (t, a) with $a - tx \in Z_U = P + J^{\mathrm{gp}}$, i.e. $\bar{a} \in t\bar{x} + \bar{P} \subseteq \bar{P}$, and $\varphi^{\mathrm{gp}}(a) \leq 0$. Since φ is tight on $\bar{P}$, $\bar{a} \in \bar{P}$ with $\varphi(\bar{a}) \leq 0$ forces $\bar{a} = 0$; then $-t\bar{x} \in \bar{P}$, and sharpness gives $\bar{x} = 0$: $x \in J$: no jump occurs, so the kernel stays J. (C2) Identical after replacing J by K: with $\ker \varphi = K$, condition (B) gives $\varphi^{\mathrm{gp}}(a) \leq 0$; write everything modulo J, with $\bar{\varphi} : \bar{P} \to \mathbb{N}$ vanishing exactly on the face K/J: (A), i.e. $a - tx \in P + K^{\mathrm{gp}}$, gives $\bar{a} \in t\bar{x} + \bar{P} + (K/J)^{\mathrm{gp}}$ after reduction; absorbing the $(K/J)^{\mathrm{gp}}$-part into $\bar{a}$ (allowed: $\bar{\varphi}$ vanishes there), (A) reads $\overline{a - tx} \in \bar{P}$, i.e. $\bar{a} \in t\bar{x} + \bar{P}$; (B) and the face property of K/J: if $\bar{\varphi}(\bar{a}) = 0$ then $\bar{a} \in K/J$ and $t\bar{x} = \bar{a} - \bar{z}$ with $\bar{z} \in \bar{P}$ is an equality inside $\bar{P}$ with sum in the face K/J, hence $t\bar{x} \in K/J$; saturation gives $\bar{x} \in K/J$, i.e. $x \in K$. If $\bar{\varphi}(\bar{a}) < 0$: $\bar{\varphi}(t\bar{x}) = \bar{\varphi}(\bar{a}) - \bar{\varphi}(\bar{z}) \leq \bar{\varphi}(\bar{a}) < 0$ while $\bar{\varphi}(t\bar{x}) \geq 0$: impossible. Conversely every $x \in K$ dies via $t = 1$, $\bar{a} = \bar{x}$ exactly as in the proof of Theorem 3.3, Claim B. The final stronger clause is the iteration of (C1) (no fold added) together with (C2) applied at the join kernel. □

Lemma 4.13 (Value group on a face quotient)**.** *Let* $K \supsetneq J$ *be a face strictly larger than* J*, and let* $\chi : \bar{P} = P/J \to \mathbb{N}$ *be tight. Then* $\bar{\chi}^{\mathrm{gp}}((K/J)^{\mathrm{gp}}) = d\mathbb{Z}$ *for some integer* $d \geq 1$ *(in particular it is nonzero).*

Proof. Were it zero, $(K/J)^{\mathrm{gp}} \subseteq \ker \bar{\chi}^{\mathrm{gp}}$; applying Proposition 2.1(c) gives $K/J = (K/J)^{\mathrm{gp}} \cap \bar{P} \subseteq \ker \bar{\chi} = \{0\}$ by tightness, hence $K = J$: contradiction. □

Lemma 4.14 (Constant strands)**.** *Let* $\mathscr{W}$ *take value* A *on every nerve piece of a cover family* $\mathcal{I}$*, with every face map* id_A*, and assume the family contains two pieces* $R_{\min} \subseteq R_{\max}$ *with* $R_{\min} \subseteq U \subseteq R_{\max}$ *for all* $U \in \mathcal{I}$*. Then the alternating Čech complex computes* $\check{H}^0 = A$ *and* $\check{H}^{q \geq 1} = 0$*.*

Proof. This is the constant-complex Lemma 4.6 applied to the family $\mathcal{I}$: $R_{\min}$ is its minimum piece, every piece receives a structure map from it, and all face maps are id_A; the conclusion is $\check{H}^0 = A$, $\check{H}^{q \geq 1} = 0$ (the $\check{H}^0$-value is read off at any piece, e.g. at $R_{\max}$). □

Lemma 4.15 (The dead level)**.** *For the dead object* $D = (P \to \{0\})$*:* $\mathcal{G}$ *is constant equal to* $F(b_P)$ *on its slice, every cover's pieces all lie over* D*, and* $H^0(D, \mathcal{G}) = F(b_P)$*,* $H^{q \geq 1}(D, \mathcal{G}) = 0$ *by Lemma* 4.14 *applied within each cover (equivalently: sheaf cohomology on a slice category with an initial object, on a constant sheaf, vanishes in positive degrees).*

Proof. Every piece over D has kernel $\supseteq \Phi_D = P$: all dead; the slice A_P/D has D initial; compatible systems are pinned by identity transitions to constants; the Čech complexes are the constant-strand ones of Lemma 4.14. □

Lemma 4.16 (Filter lemma)**.** *Let* Ξ *be a finite poset of levels with a unique largest element* $\top$*, and let* R *be a cover of* X *with a presheaf* $\mathscr{W}$ *whose pieces carry order-respecting* Ξ*-labels—every structure map* $\overline{\tau} \to \overline{\tau'}$ *has* $\mathrm{lev}(\overline{\tau}) \leq^{\Xi} \mathrm{lev}(\overline{\tau'})$*, so by Lemma 4.8 each face of a nerve piece has label* $\leq^{\Xi}$ *the piece's label. Suppose: (F1) the top level* $\top$ *is a differential-closed constant strand with a minimum piece (Lemma 4.6):* $\mathscr{W}$ *is constant equal to a single* A *on* $\top$*-nerve pieces, all structure maps among them inducing* id_A*, and a fixed piece* $H_\top$ *maps to every* $\top$*-piece; (F2) every nerve piece labelled* $\top$ *arises by adjoining to a lower-labelled tuple* τ *a piece* U *whose structure map factors through* $H_\top$*, with* $\overline{(U, \tau)}$ *again* $\top$*-labelled; the piece* U *is a fold whose structure map's image*

lies in the kernel of the transport $\mathscr{W}(\overline{\tau} \cup U) \to \mathscr{W}(\overline{\tau})$ (*the* dead-adjunction hypothesis). *If* z *is a degree-*$q \geq 1$ *cocycle whose value at every* $\top$*-labelled nerve piece vanishes, then* z *is cohomologous, within the same cover, to a cocycle supported on labels* $<^{\Xi} \top$ (*the next-level values being the signed transports of Lemma 4.7); iterating down the finite poset,* z *pulls* (*after no refinement: at the level of cocycles*) *to a cocycle supported at the initial level.*

Proof. Define, for a lower tuple τ (all labels $<^{\Xi} \top$), a cochain h on $\top$-pieces by $(he)_{(U,\tau)} := e_\tau$ when (F2) applies (transported: e_τ lies at the lower level; the value is its image at $\overline{(U,\tau)}$, well-defined since the map $\overline{(U,\tau)} \to \overline{\tau}$ has constant transport by (F2)'s factoring through $H_\top$). Then the cocycle equation $(\partial z)_{(U,\tau)} = 0$ reads, isolating the face omitting U,

$$z_\tau \text{ (transported)} \;=\; \sum_i (-1)^{i+1}\, \mathscr{W}(d_i)\big(z_{(U,\partial_i\tau)}\big) \quad \text{on } \top\text{-pieces},$$

whose right side is $(\partial h z)_{(U,\tau)}$: precisely the constant-complex contraction of Lemma 4.6 run against the anchor $H_\top$. Subtracting $\partial(hz)$—a coboundary supported on $\top$ by (F1)—removes the $\top$-component, and by Lemma 4.7 the cocycle is now supported one level lower, the next-level values shifted by the signed transport. Finiteness of Ξ terminates the descent at the initial level. □

Lemma 4.17 (Sieve: fold terraces)**.** *Let* R *be a standard rigid cover of* $i(b)$ *and* z *a* $\mathcal{G}$*-Čech cocycle of degree* $q \geq 1$ *on* R*. Then* z *is cohomologous within* R (*no refinement*) *to a cocycle vanishing at every nerve piece containing a kernel-*J *fold* (*the dead sector is left arbitrary*)*.*

Proof. A kernel-J fold is a tight functional $\varphi : \bar{P} \to \mathbb{N}$; up to rational multiple it is a point of the pointed, strictly convex rational polyhedral cone $\mathrm{Cone}(\bar{P})^\vee$, so there are finitely many fold *rays* in R: enumerate them $\mathfrak{f}_1, \ldots, \mathfrak{f}_m$. For a ray index r let the *r-fold strand* consist of the tuples containing $\mathfrak{f}_r$ but no fold of a different ray. On such a tuple $\tau = (\mathfrak{f}_r, \sigma)$ every position of σ is iB or a fold on ray r, hence has kernel $\subseteq J$; by Lemma 4.11(i) Case A and Lemma 4.12(C1) the overlap $\overline{\tau} = \mathfrak{f}_r \otimes \overline{\sigma}$ still has kernel J, so all structure maps among these nerve pieces induce the identity on the $\mathcal{G}$-value $F(b)$. Define a $(q-1)$-cochain w_r on the strand by

$$(w_r)_\sigma \;:=\; z_{(\mathfrak{f}_r,\sigma)} \qquad (\text{taken } = 0 \text{ if } (\mathfrak{f}_r, \sigma) \text{ has a repetition}),$$

which is well-defined because adjoining $\mathfrak{f}_r$ keeps the kernel J (above). Then for a strand cocycle z, $(\partial w_r)_\tau = z_\tau - (w_r\partial)_\tau$: this is the constant-strand identity, since with all transports $\mathrm{id}_{F(b)}$

$$(\partial w_r)_\tau = \sum_i (-1)^i z_{(\mathfrak{f}_r,\partial_i\tau)}, \qquad (w_r\partial)_\tau = z_\tau - \sum_i (-1)^i z_{(\mathfrak{f}_r,\partial_i\tau)},$$

so $(\partial w_r + w_r\partial) = z$ on the strand: the strand is acyclic in degrees ≥ 1 and the r-component of z is ∂w_r. The strands for distinct r are disjoint (a tuple with two distinct-ray folds lies in none) and their union is differential-closed, so by repeated application of the exact-subquotient Lemma 4.7 the cocycle $z - \partial(\sum_r w_r)$ is supported on tuples containing *no* kernel-J fold, as claimed. The dead tuples are never touched. □

Lemma 4.18 (Common refinement of covers)**.** *Let* $R = \{U_\alpha\}$ *and* $S = \{V_\beta\}$ *be covers of* $i(b)$ *in the sliced birhizomic site. Then* $R^\flat := \{U_\alpha \otimes_{i(b)} V_\beta\}_{\alpha,\beta}$ *is a cover of* $i(b)$ *refining both* R *and* S (*via the structure maps to* U_α *and to* V_β)*; in particular the Čech colimit of Remark 4.3 is directed.*

Proof. The birhizomic topology is a pretopology: its coverings are stable under pullback and composition (Remark 2.13; the pushouts exist by Construction 2.2). For each fixed U_α the family $\{U_\alpha \otimes_{i(b)} V_\beta\}_\beta$ is the pullback of the cover S along $U_\alpha \to i(b)$, hence a cover of U_α; composing these over the cover $R = \{U_\alpha\}$ of $i(b)$, the family $\{U_\alpha \otimes_{i(b)} V_\beta\}_{\alpha,\beta}$ covers $i(b)$. It refines R and S via the two structure maps $U_\alpha \otimes V_\beta \to U_\alpha$ and $U_\alpha \otimes V_\beta \to V_\beta$ given by the pushout universal property. □

Proposition 4.19. *(a)* $H^0(i(b), \mathcal{G}) = F(b)$. *(b) For* $q \geq 1$*, every class* $[z] \in \check{H}^q(R; \mathcal{G})$ *on a standard rigid cover is, after a common refinement, cohomologous to a cocycle that (i) has coboundary values with coefficients in* $F(b_P)$ *at every nerve piece mapping to* D*, and (ii) vanishes at every nerve piece containing a kernel-*J *fold. The final vanishing* $H^{q>0}(i(b), \mathcal{G}) = 0$ *is* not *proved for arbitrary* F *(see the Closing remark).*

Proof. By the comparison theorem (Remark 4.3), $H^q(i(b), \mathcal{G}) = \operatorname{colim}_R \check{H}^q(R; \mathcal{G})$ over the nerves of Definition 4.1, the colimit taken over the standard rigid covers (cofinal by Definition 4.10), which form a directed system by the common-refinement Lemma 4.18. We prove $\check{H}^0(i(b), \mathcal{G}) = F(b)$—true already on the nose: $\mathcal{G}$ is a sheaf (Theorem 3.3) and $\mathcal{G}(i(b)) = F(b)$ (Proposition 3.1)—and that every class $[z] \in \check{H}^{q\geq 1}(R; \mathcal{G})$ pulls to zero on a further refinement. We work one degree $q \geq 1$ at a time, associating to a q-cocycle its $(q-1)$-*faces* (the values $z_{(U_0,U_1)}$), and use the sheaf axiom at each step to descend. For $\bar{P} = \{0\}$ the slice is the dead level and Lemma 4.15 gives the vanishing; for $\dim \bar{P} \leq 1$ the whole nerve of a standard rigid cover is the constant strand with minimum iB, so Lemma 4.14 ($R_{\min} = R_{\max} = iB$) gives $\check{H}^{q\geq 1} = 0$ directly (these are the base cases). Assume now $\dim \bar{P} \geq 2$.

Step 1 (dead-sector hygiene). Let R be a standard rigid cover. The sifted family $S := \{U \otimes_{i(b)} D : U \in R\}$ is the pullback of R along $D \to i(b)$, hence a cover of D (Lemma 4.18); pass to the common refinement of R and S (Lemma 4.18 again) and keep the notation R, z. Every nerve piece of S is dead, and on the slice over D the sheaf $\mathcal{G}$ is the constant sheaf $F(b_P)$ (Lemma 4.15); by the constant-complex Lemma 4.6 with anchor $H = D$, the restricted cocycle is a coboundary: $z|_S = \partial w$ for a $(q-1)$-cochain w on S. Pulling w back to R along the refinement map, we may subtract ∂w from z; hence after this refinement

the value of z at every nerve piece that maps to D is the value of a coboundary with coefficients in $F(b_P)$. (∗)

We assert nothing further about the dead sector; we do not pin it to a single constant.

Step 2 (fold terraces). Apply Lemma 4.17 to z: within the same cover it is cohomologous to a cocycle vanishing at every nerve piece containing a kernel-J fold; replace z by this representative. In particular $z_{\mathfrak{f}} = 0$ for every kernel-J fold $\mathfrak{f}$ and $z_\tau = 0$ at every tuple containing one. (The identification $\bar{\chi}^{\mathrm{gp}}((K/J)^{\mathrm{gp}}) = d\mathbb{Z}$, $d \geq 1$, used in the kernel computations is Lemma 4.13.)

Step 3 (where the argument stops). After Step 2, z vanishes on every tuple containing a kernel-J fold; the surviving tuples are built from iB and the CFA pieces $i(P/K)$. The natural candidate to conclude $[z] = 0$ is the iB-contraction: adjoining iB is internal on a fold-free tuple, and $(hz)_\tau := z_{(iB,\tau)}$ ($= 0$ on repetition) is well-defined with $(\partial hz)_\tau = z_\tau - (h\partial z)_\tau$, as in Lemma 4.17. However, to have $z_\tau = (\partial hz)_\tau$ at a fold-free tuple one uses $\partial z = 0$ there, and the residual term of $\partial(hz)$ at the dead adjacent tuples must then cancel the contraction: this requires $\mathcal{G}$ to agree on the base face and its quotient faces, i.e. the transitions $\mathcal{G}(i(b)) \to \mathcal{G}(i(P/K))$ to be identities. For a general presheaf F these are the quotient maps $F(b) \to F(B(i(P/K)))$, which need not be identities, and the discrepancy cannot be absorbed into Step 1's w (which lives only on the dead level). The vanishing follows in those special cases where $\mathcal{G}$ agrees on base and quotient faces, but not for arbitrary F; the document therefore does *not* assert $H^{q>0}(i(b), \mathcal{G}) = 0$ in general. The degree-0 part is Theorem 3.3 and Proposition 3.1. □

We close by recording exactly what has been proved and what remains open, in accordance with the requirement never to present conjecture as fact.

Proposition 4.20 (The sheaf and the degree-zero unit)**.** *For every finitely generated monoid P and every presheaf F of abelian groups on B_P^{op}: (i) $\pi^{-1}F = G_F$ is a sheaf (Theorem 3.3); (ii) the unit $F \to \pi_*\pi^{-1}F$ is an isomorphism, i.e. $\mathrm{R}^0\pi_*(\pi^{-1}F) = F$.*

Proof. (i) is Theorem 3.3. (ii): with the trivial topology on B_P^{op} sheaves equal presheaves, and $H^0(i(b), \mathcal{G}) = \mathcal{G}(i(b)) = F(b)$ by Theorem 3.3 and Proposition 3.1, so the degree-zero unit $F \to \pi_*\pi^{-1}F$ is the identity sheaf map. □

Closing remark: higher direct images and the general case. Two points are deliberately left open, and this note does not decide them. (1) *Higher direct images.* Proposition 4.19 reduces $H^{q>0}(i(b), \mathcal{G})$, after the fold terraces are killed (Lemma 4.17) and the dead sector is made coboundary-valued, to an iB-contraction on the residuum. That contraction requires $\mathcal{G}$ to agree on the base face and its quotient faces; for an *arbitrary* presheaf F the structure map $\mathcal{G}(i(b)) \to \mathcal{G}(i(P/K))$ is the quotient map $F(b) \to F(B(i(P/K)))$, which need not be the identity, so the residuum argument does not close $H^{q>0}(i(b), \mathcal{G}) = 0$ for all F, and the document does not claim it does. Consequently the full quasi-isomorphism asked in the problem ($\mathrm{R}^{q>0}\pi_*(\pi^{-1}F) = 0$ for $q > 0$) is *not proved here.* (2) *Non-finitely generated P.* As recorded in the Scope hypothesis (§1) and Remark 2.4, without finite generation of P the map $P \to P/K$ need not be finitely presented and the face poset can be infinite, so the birhizomic-cover machinery and the finiteness used in the sieve break; the general case is obstructed exactly there and is likewise left open. No counterexample to either open point is known to this note: as stated, the problem is *open*, with the degree-zero part and the sheaf property established above for finitely generated P.